

MAZEN WAEI

Software Engineer | Backend & Full-Stack | AI/ML

+20 128 608 0266 • mazenwael575@gmail.com • Cairo, Egypt • [LinkedIn](#) • [GitHub](#)

EDUCATION

B.Sc. Software Engineering — Computer Science

Cairo University

Oct 2022 – Jun 2026

Cairo, Egypt

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Database Systems, Object-Oriented Programming, Computer Networks

TECHNICAL SKILLS

Languages: Python, Java, C#, C, C++, JavaScript (ES6+), TypeScript

Web & Frameworks: Flask, ASP.NET Core, .NET, REST API Design, WebSockets, HTML5, CSS3

Databases: MySQL, SQL Server, Query Optimization, Schema Design, EF Core

DevOps & Tools: Git, GitHub, Docker, Nginx, GitHub Actions (CI/CD), Linux, Shell Scripting

AI & ML: Deep Learning, CNNs, RNNs, LSTMs, Transformers, TensorFlow, scikit-learn, RAG, LangChain, Sentence Transformers, Gemini API, Vector Embeddings, Pandas, Seaborn, Matplotlib

Concepts: OOP, Design Patterns, SOLID Principles, Unit Testing, Agile, Data Structures & Algorithms

EXPERIENCE

Software Engineering Intern

ALX Africa

Sep 2023 – Nov 2024

Cairo, Egypt

- Led a cross-functional 3-engineer team to architect and ship MarketMate, a production-grade full-stack marketplace platform, delivering the project on schedule across a 14-month engagement.
- Engineered RESTful API endpoints across 8 resource types in Flask, applying MySQL query indexing, N+1 query elimination, and endpoint-level caching — reducing average API response time by 40%.
- Automated the full build-and-deploy pipeline using GitHub Actions CI/CD, cutting manual deployment steps from 12 to 3 and decreasing total deployment time by over 60%.
- Deployed Flask applications to production Linux servers using Nginx as a reverse proxy with Docker containerization, ensuring zero-downtime releases and environment parity across dev/staging/production.
- Architected normalised MySQL database schemas for a product catalog supporting 50+ SKUs, implementing indexing strategies that sustained sub-100ms query performance under concurrent load.

PROJECTS

MarketMate — Full-Stack Marketplace Platform [Github](#) Flask · MySQL · Docker · GitHub Actions

- Designed and built a full-stack marketplace with JWT-based authentication, a product catalog serving 50+ SKUs, and real-time buyer/seller notifications via WebSockets; deployed on Linux with Nginx and Docker.
- Reduced API response latency by 30% through systematic MySQL indexing and N+1 query elimination; improved system reliability with a CI/CD pipeline that cut deployment overhead by 60%.

Egyptian Legal AI — RAG-based Legal Assistant (Graduation Project) [Github](#) Python · FastAPI · LangChain · Gemini · React

- Architected a Retrieval-Augmented Generation (RAG) system for Egyptian Labour Law, delivering accurate Arabic legal responses with cited article references using FAISS vector search, LangChain agents, and Gemini 2.0 — achieving 95.7% accuracy across 15 legal test cases.
- Built a production FastAPI backend with SQL Server chat history and JWT authentication, handling concurrent sessions with sub-5s response latency and a 100% pass rate across 75 comprehensive test cases.

Airbnb Clone — Distributed Booking API [Github](#) Flask · MySQL · HTML/CSS

- Built a RESTful API in Flask handling booking and listing workflows across 5 normalised resource models; implemented full CRUD operations with a responsive HTML/CSS frontend and SQL schema enforcement.

Heart Disease Prediction — ML Classification Model [Github](#) Python · scikit-learn · Matplotlib · Seaborn

- Trained Logistic Regression and Random Forest classifiers on the UCI Heart Disease dataset; achieved 87% classification accuracy after hyperparameter tuning with GridSearchCV.
- Conducted exploratory data analysis across 13 clinical features and produced 8 diagnostic visualisations to surface feature correlations and class imbalances.

Habit Tracker App — ASP.NET Core CRUD Application [Github](#) C# · ASP.NET Core · SQL Server · EF Core

- Designed EF Core database models and implemented full CRUD controllers with Razor views in ASP.NET Core MVC, delivering a functional personal productivity tool with persistent SQL Server storage.

CERTIFICATIONS

Full Stack Software Engineering <i>ALX Africa</i>	Sep 2023 – Nov 2024
Deep Learning Specialization <i>DeepLearning.AI (Coursera)</i>	Jun 2025 – Oct 2025
AI & Machine Learning Program <i>Microsoft / Sprints</i>	Jun 2025 – Nov 2025
EF SET English Certificate — C2 Proficient (71/100) <i>EF SET</i>	Mar 2025